

Trevor Crouse

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EDUCATION

Cornell University, Duffield College of Engineering **Ithaca, New York**
Bachelor of Science in Mechanical Engineering *Expected Graduation May 2028*

- **Relevant Coursework:** Dynamics, Statics and Mechanics of Solids, Thermodynamics, Mechanical Design, Electromagnetism, Waves and Quantum Physics, Differential Equations, Linear Algebra
- **Campus Involvement:** Design Build Fly Project Team, Zeta Psi Fraternity, Club Hockey, Makers Club, American Society of Mechanical Engineers, Aerodynamic Research

RELEVANT EXPERIENCE

Design, Build, Fly Project Team **Ithaca, New York**
Aerodynamics and Controls Sub-team Lead *November 2024 – Present*

- Directed three work sessions a week of a 14-member aerodynamic team, creating all components of the wing and tail on a high-performance RC aircraft, utilizing SolidWorks CAD for design and ANSYS Fluent for aerodynamic performance
- Collaborated with cross-functional sub-team to design and integrate subsystems, using ANSYS Mechanical FEA to perform stress analysis on components and to create simulations of ground competition missions
- Led technical documentation on a 58-page competition report for the annual AIAA Design, Build, Fly competition and competed in Wichita, Kansas against over 100 different U.S. and international universities

Zeta Psi Fraternity **Ithaca, New York**
Executive Board Position for Event Planning and Risk Management *April 2025 – Present*

- Managed and allocated a \$30,000 budget for events, including catering, venue selection, and vendor coordination
- Developed and implemented risk management strategies, including transportation and safety protocols for over 90 people
- Coordinated logistics with internal committees and external partners to ensure smooth execution of large-scale events

PROJECTS

Aircraft Wing Design and Manufacturing **Ithaca, New York**
Project for Design, Build, Fly *August 2025 – Present*

- Optimized airfoil and wing shape around AIAA constraints using XFLR5 and a self-made MATLAB sizing script
- Designed a manufacturable full-scale wing in SolidWorks creating useful assemblies to test aileron deflections and to use in ANSYS CFD to perform lift and drag analysis
- Manufactured the wing, iterating five different versions reducing mass and decreasing induced drag using hardwood, balsa, and 3D designed integration pieces

BuzzKill: Mechanical Control Product for an Invasive Insect **Ithaca, New York**
Group Project for MAE 2250 – Mechanical Design *January 2026 – Present*

- Designed a mechanical solution to control spotted lanternfly populations in vineyards, commissioned by E. & J. Gallo Winery and National Grape
- Tested various solutions, deciding on a jaw bucket design attached to an extendable rod using high mechanical advantage to reduce labor intensity, creating a scalable and cost-efficient solution
- Selected high strength, weather resistant materials and applied DFM principles to optimize the jaw bucket for field durability and low-cost manufacturing

Wing Endplate Aerodynamic Research **Ithaca, New York**
Independent Project *January 2026 – February 2026*

- Conducted research on wing endplates efficiency and contribution to lift at low Reynolds Numbers, using professor insight to design endplate shapes in SolidWorks and using ANSYS CFD to perform lift-drag analysis
- Wrote a report on the effectiveness of wing endplates, balancing their effectiveness while comparing their benefits with weight and manufacturing constraints

CERTIFICATIONS

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- AutoCAD Basic Training | Issued January 2026 (Udemy)
 - Aerospace Engineering: Airplane Design and Aerodynamic Basic | Issued October 2024 (Udemy)
 - Automotive Engineering: Automobile Fundamentals and Advanced | Issued October 2024 (ValueKnow)

SKILLS

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- Skills: ANSYS Fluent, CFD, SolidWorks CAD, Fusion 360, MathWorks MATLAB, Python, Aerodynamic Analysis, Mechanical Design, Machine Shop Manufacturing, Microsoft Office, Excel, Technical Documentation